

PresCoach

CTO-to-CTO architecture review

An AWS-native agentic presentation coaching platform built with Kiro and AI-DLC practices.

The punchline: this is not just a demo. It is a useful pattern for taking agentic systems from idea to governed software delivery without pretending the robots can skip architecture review. Nice try, robots.

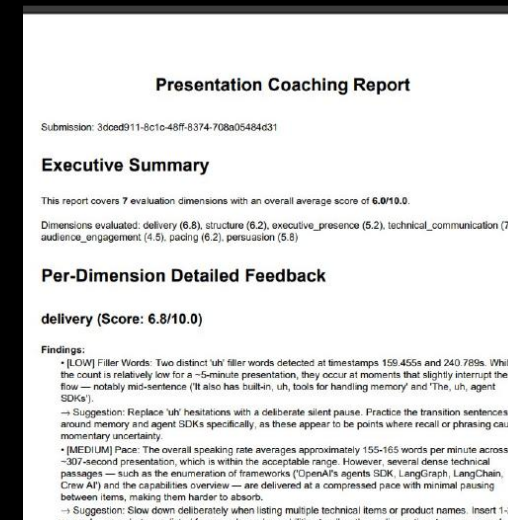
Dark theme retained from the working deck



AI-Powered Presentation Coaching

Upload your presentation audio and receive detailed feedback across 7 dimensions: delivery, structure, executive presence, technical communication, audience engagement, pacing, and persuasion.

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**AWS-native
serverless +
agentic**

**Built for
zero idle cost**

EXECUTIVE LENS

The CTO readout

PresCoach is a working AWS-native product pattern, not a lab toy. It accepts audio, prepares it deterministically, then hands it to independent AI evaluators that produce a coaching report.

7

AI coaching dimensions

20

AWS services used

726

tests in all workflows

\$33.69Build spend for
Environment and AI Tool

\$6.59 AWS + \$27.10 Kiro

**Architecture decision
worth a CTO review****Where to harden
before real users****How to scale
without cost traps****What AI-DLC
actually changed**

WHAT IT DOES

The product loop is simple, the system behind it is not

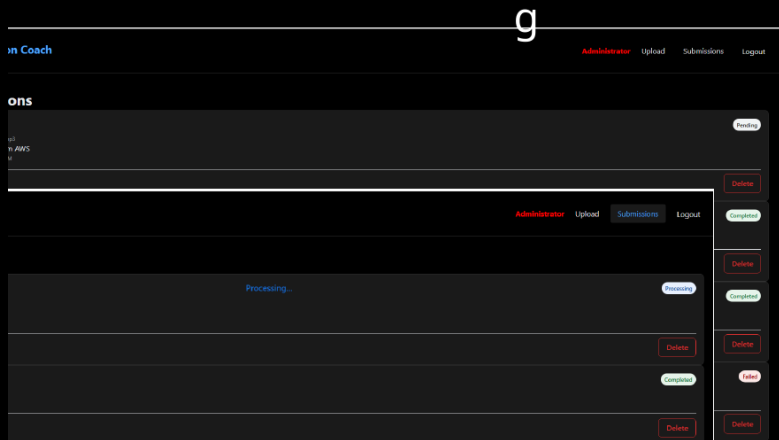


Upload
mp3, wav, m4a, aac

Prepare
transcribe + chunk +
embed

Evaluate with
seven agent lenses

Report
PDF coaching
output



User-facing flow

The deliberate split: deterministic processing does the mechanical work, while the agentic layer handles judgment. That separation is what makes the design easier to govern.

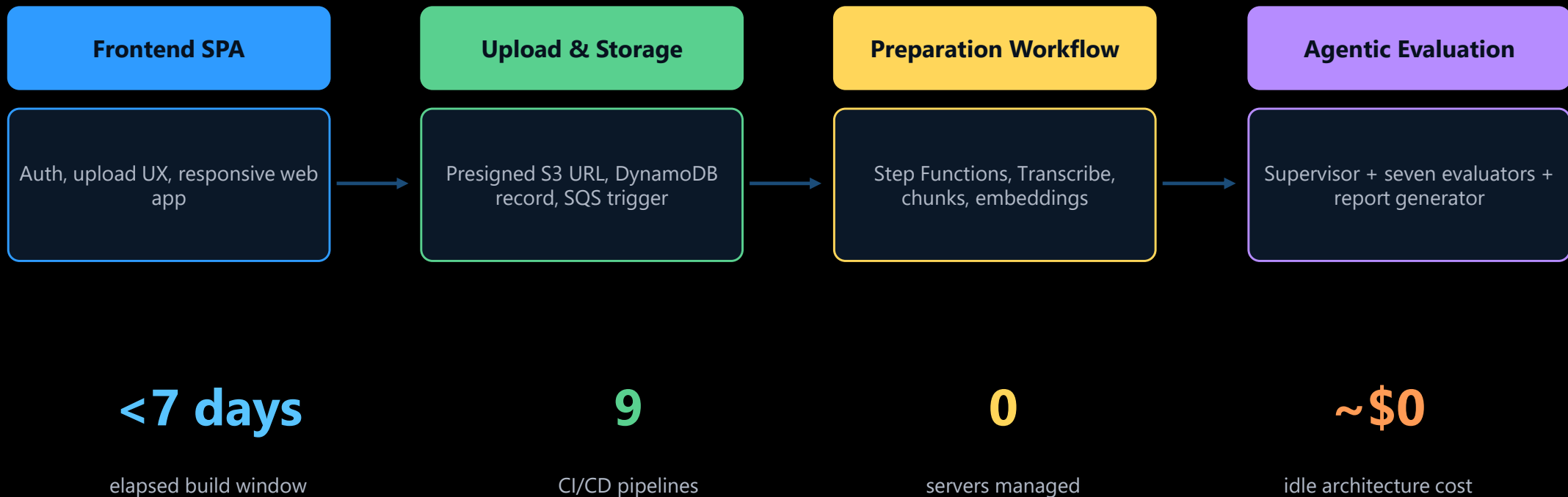
Video is recognized but feature-flagged off in the MVP

Accepted audio formats and video flag behavior come from the end-to-end processing flow.

DELIVERY MODEL

Build approach: AI-DLC turned the idea into deployable work units

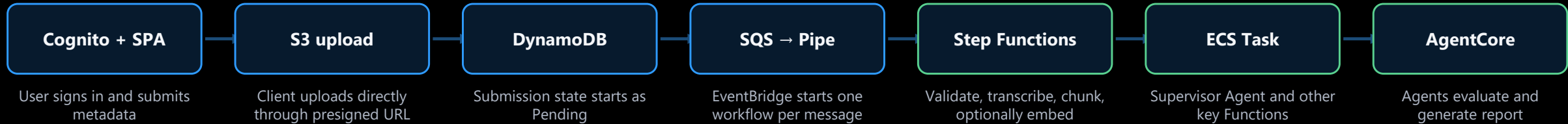
The useful lesson is not that code was generated quickly. The useful lesson is that the work was decomposed into independently testable “bolts” with clear contracts.



Retrospective describes four work units, bolt-based interfaces, CI/CD pipelines, and zero-idle service selection.

RUNTIME FLOW

End-to-end processing is event-driven and inspectable

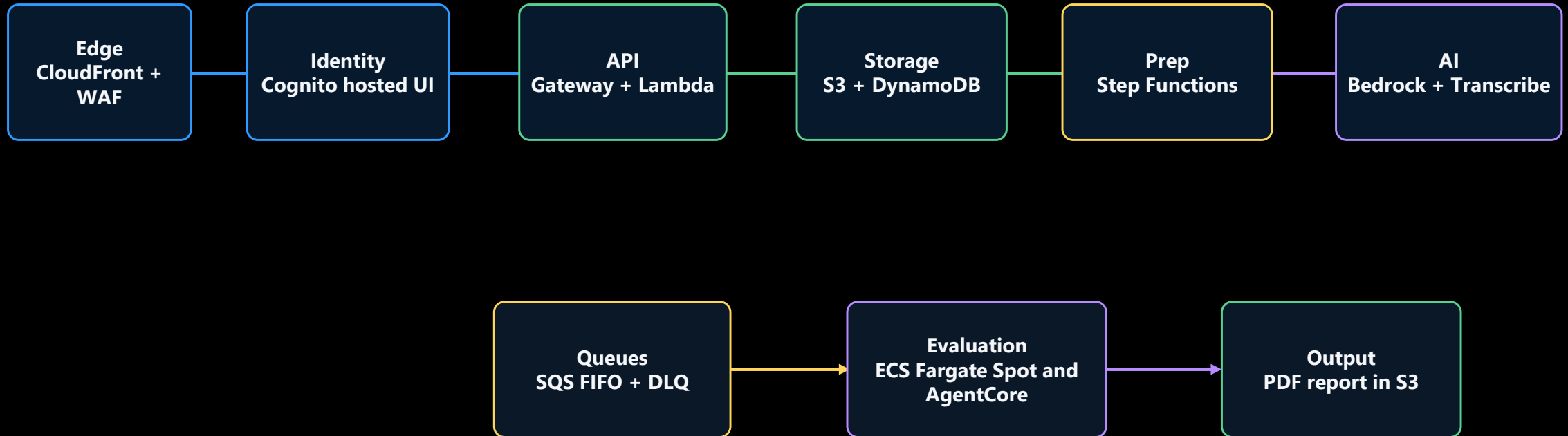


The handoff contract matters more than the tool choice. The upload service does not know how evaluation works. The evaluator does not care how the file arrived. That keeps each boundary testable and replaceable.

State transitions
Pending → **Processing** → **Completed**
or **Failed**

Failure path
retries → **HandleFailure** → **SNS + DLQ**

Architecture, viewed like a CTO would review it



Why this is a good MVP architecture

- Files never pass through Lambda
- Async processing gives back-pressure
- Feature flags allow runtime tuning
- Least-privilege IAM can be reviewed per work unit

What I would challenge before production

- Model quotas and throttling behavior
- Regional failure stance and user-pool strategy
- PII retention and report lifecycle policy
- Evaluation result explainability and traceability

Architecture summarizes services described in project overview, flow, and retrospective.

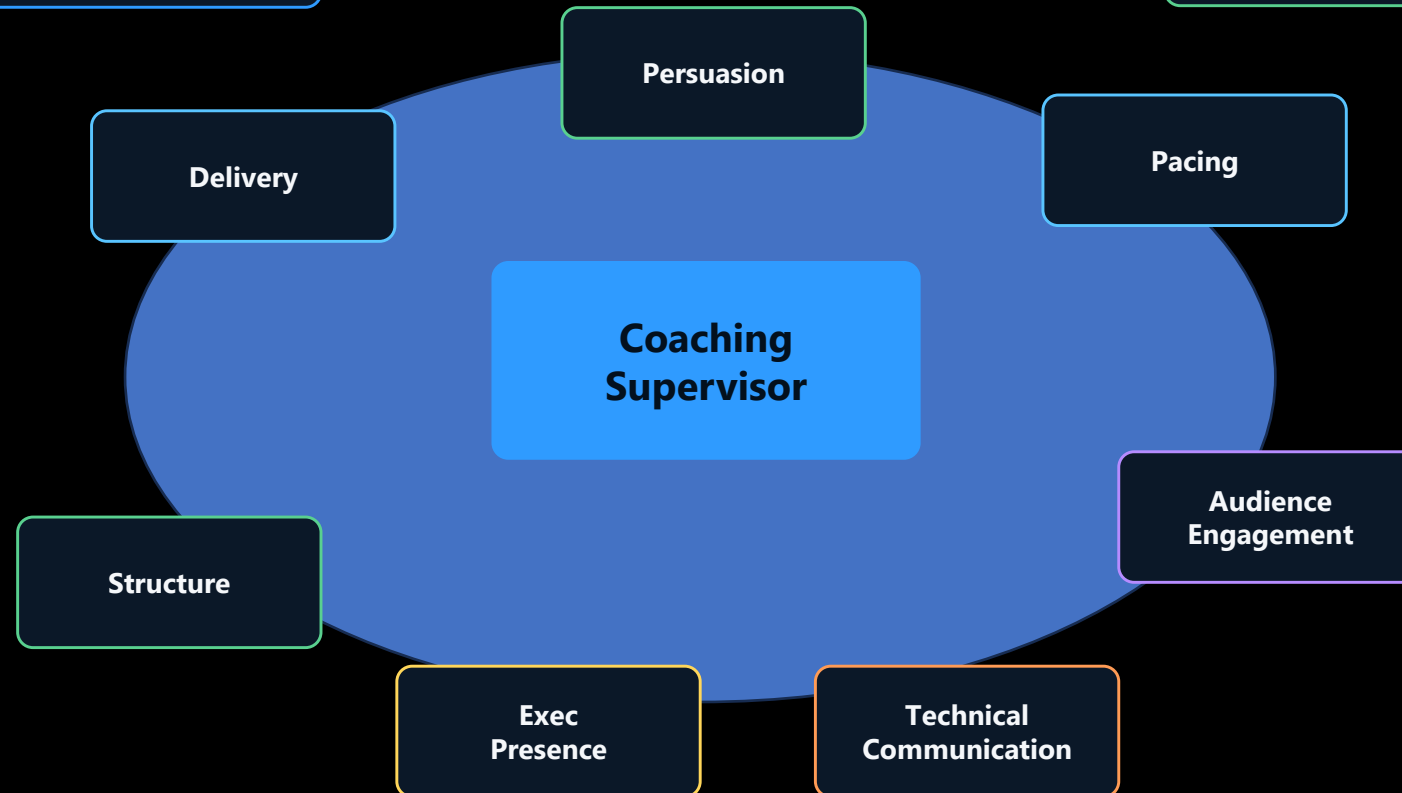
AI EVALUATION

The agentic layer is composable by design

This is the part that justifies an agentic approach. The value comes from multiple independent lenses reviewing the same asset, not from one big prompt pretending to be a committee.

Agent definitions live in a JSON manifest and can be enabled or disabled without code changes.

The report generator assembles structured results into a 10-13 page PDF coaching report.



Seven dimensions and JSON manifest behavior come from project overview and source deck.

COST AND SCALE

The economics are exactly why serverless fits

For spiky, early-stage usage, the cost goal is not “cheapest at theoretical peak.” It is “do not pay for infrastructure while product-market fit is still loading.”

Demo / PoC

10/month

 **\$0.55****Regular use**

500/month

 **\$28****Conference spike**

10,000/week

 **\$531****Quiet after spike**

following month

 **\$2.35**

**Bedrock dominates
the bill at scale**

**Zero idle
for Lambda, API Gateway,
Step Functions, SQS, SNS,
Bedrock**

**S3 vector store
keeps MVP cost near zero**

Cost scenarios and zero-idle principle come from project retrospective cost analysis.

COST DECISION

S3 as the first vector store is the right kind of cheap

This is one of those choices where CTO discipline matters. For the MVP, S3 is not the “best vector database.” It is the best way to avoid paying for one before the workload proves it deserves one.

S3 JSON embeddings

Idle cost

\$0 storage only

Read pattern

Per-submission retrieval

Latency

~50-100 ms per file

Move when

MVP and variable load

OpenSearch Serverless / pgvector

~\$175/month minimum

Similarity search across corpora

~10 ms query path

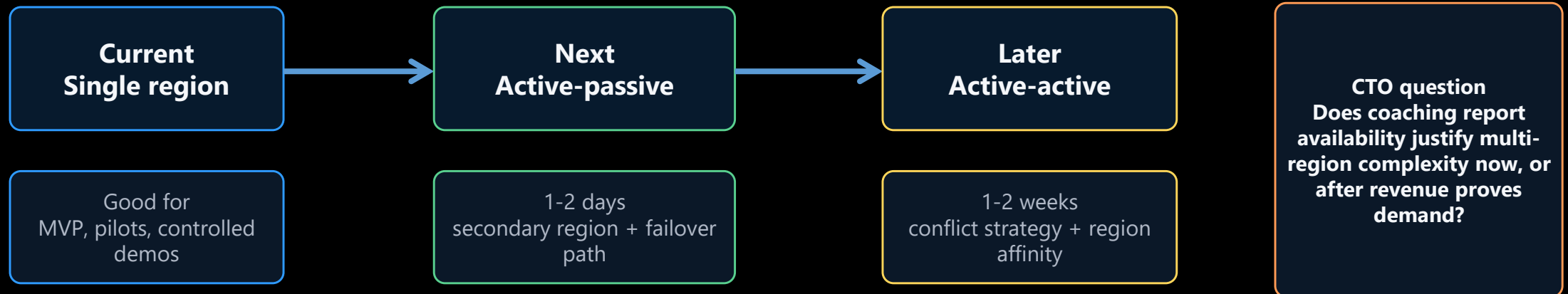
>50K presentations/month or cross-submission search

Design win: vector-store-type and vector-store-endpoint are SSM parameters, so the upgrade path is operationally clean.

RESILIENCE

Operational resilience is solid for MVP, but the regional stance is a decision

Current state is single-region in us-east-1 but multi-AZ compatible by design and construction. Single Region is acceptable for an MVP, but production requires an explicit answer to how much regional failure the business will tolerate.



Low-friction upgrades: DynamoDB Global Tables, S3 CRR, second regional API Gateway, replicated Step Functions and Lambda stacks, CloudFront origin routing.

RISK REGISTER

The production-hardening backlog is manageable, not mysterious

The system is deployed and the end-to-end pipeline runs. The remaining work is application-level correctness plus the usual enterprise-grade hardening. Good news: no smoke machine required.

Known app fixes

Vector retrieval uses the wrong API. Supervisor results need parser handling. My Submissions page throws a 500 after report creation.

Security

Token storage posture, signed URL TTLs, WAF rules, IAM boundaries, data retention, PII handling.

Reliability

Bedrock throttling policy, ECS Spot interruption handling, DLQ runbooks, retry budgets, idempotency.

Observability

Trace a submission across Cognito, API, S3, Step Functions, queues, ECS, Bedrock, and report generation.

FinOps

Budgets, Bedrock spend alarms, lifecycle policies, model-choice controls, chargeback by user or tenant.

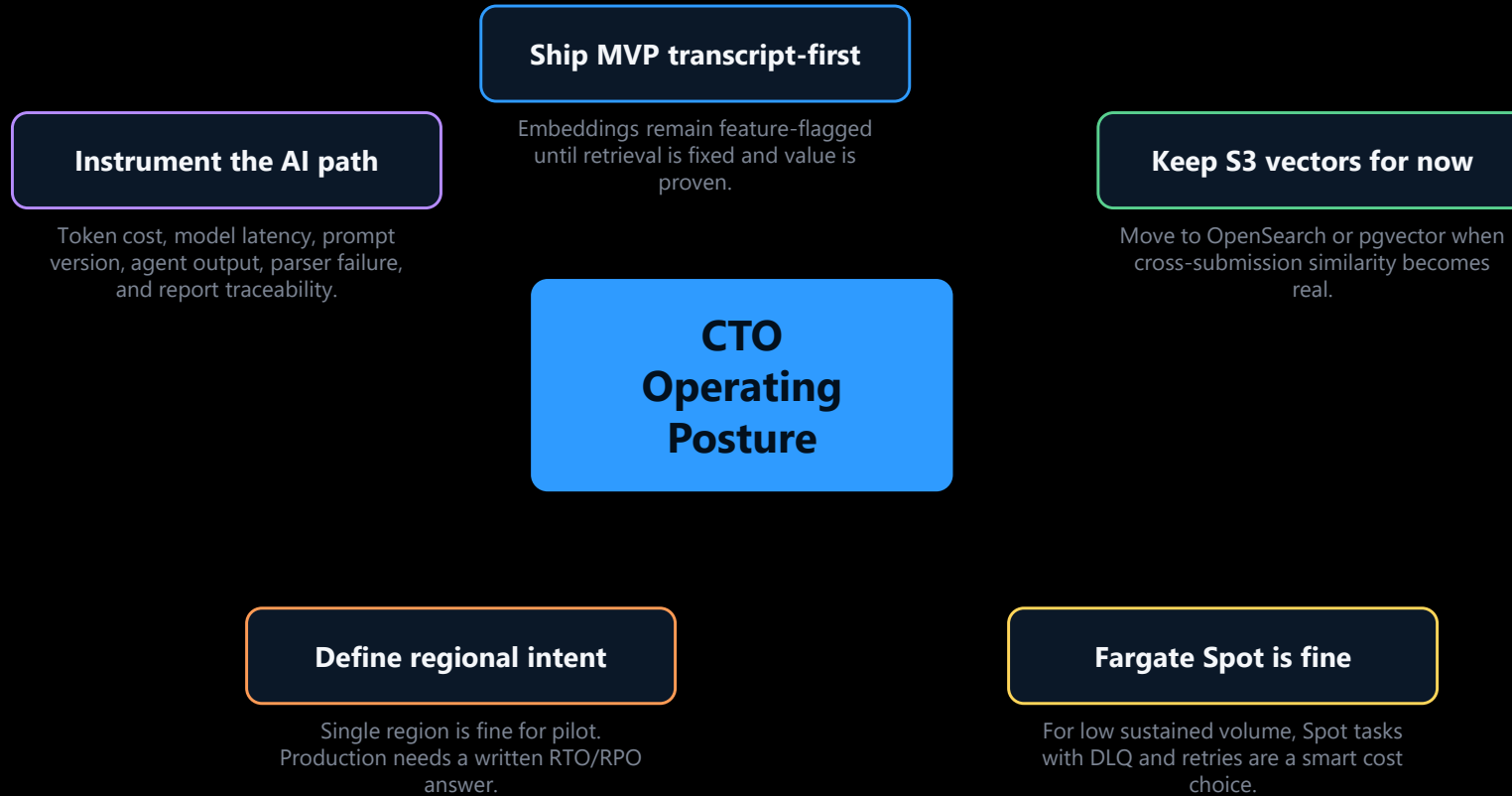
Priority one
Fix parsing and retrieval first. Then run a thin production-readiness review before adding more agents.

Known application issues come from project overview and source deck. Hardening items are CTO recommendations derived from the architecture.

WHERE I WOULD SPEND TIME

CTO decision radar

The architecture is good enough to advance. The discussion should move from “can this work?” to “what operating posture do we want?”



This slide distills CTO decisions from the supplied architecture, flow, cost, and current-state documents.

NEXT MOVES

A practical 30/60/90 day path

Do not boil the architecture ocean. Fix the known seams, prove the evaluator quality, then harden only the parts that have a user or revenue reason to exist.

30 days

Stabilize MVP

- Fix vector retrieval and parser handling
- Remove My Submissions 500 error
- Add end-to-end trace ID
- Budget alarms on Bedrock usage

60 days

Prove quality

- Golden set for coaching quality
- Prompt and model versioning
- Agent output schema enforcement
- Report scoring calibration

90 days

Prepare production posture

- Retention and PII controls
- RTO/RPO decision
- Quotas and load test
- Tenant or user-level cost attribution

Exit criteria

A CTO should be able to trace one submission from upload to report, explain one agent's score, and predict the cost of the next 1,000 runs.

CLOSE

The CTO-to-CTO takeaway

PresCoach is a strong pattern because it combines three things that usually get separated: a useful product experience, an AWS-native cost model, and an AI-DLC delivery method that keeps architecture boundaries visible.

What is working

Clear work-unit boundaries, async flow, managed services, zero idle cost, composable agents.

What to fix

Retrieval, parser handling, status page behavior, traceability, hardening of AI and data controls.

What to decide

Regional posture, vector store timing, quality gates, model spend controls, production-readiness threshold.

**Bottom line: advance it, but make the next phase boring in the best possible enterprise way.
Boring is where systems go to become dependable.**

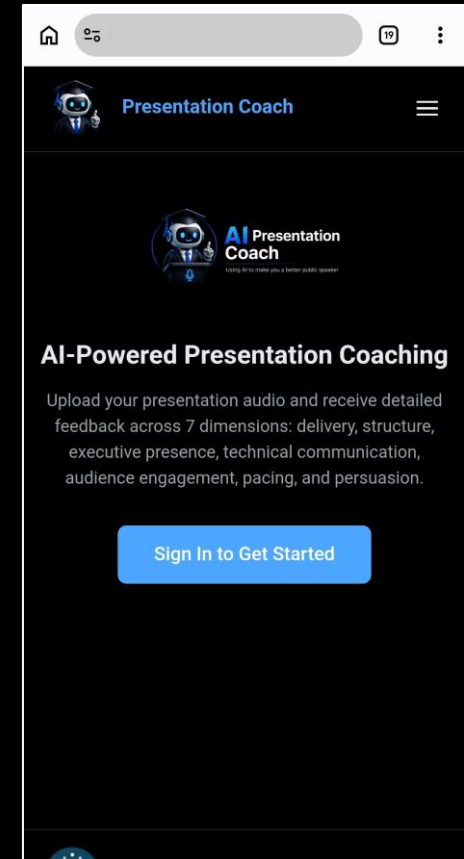
Appendix Material

The Web App

Responsive Web App

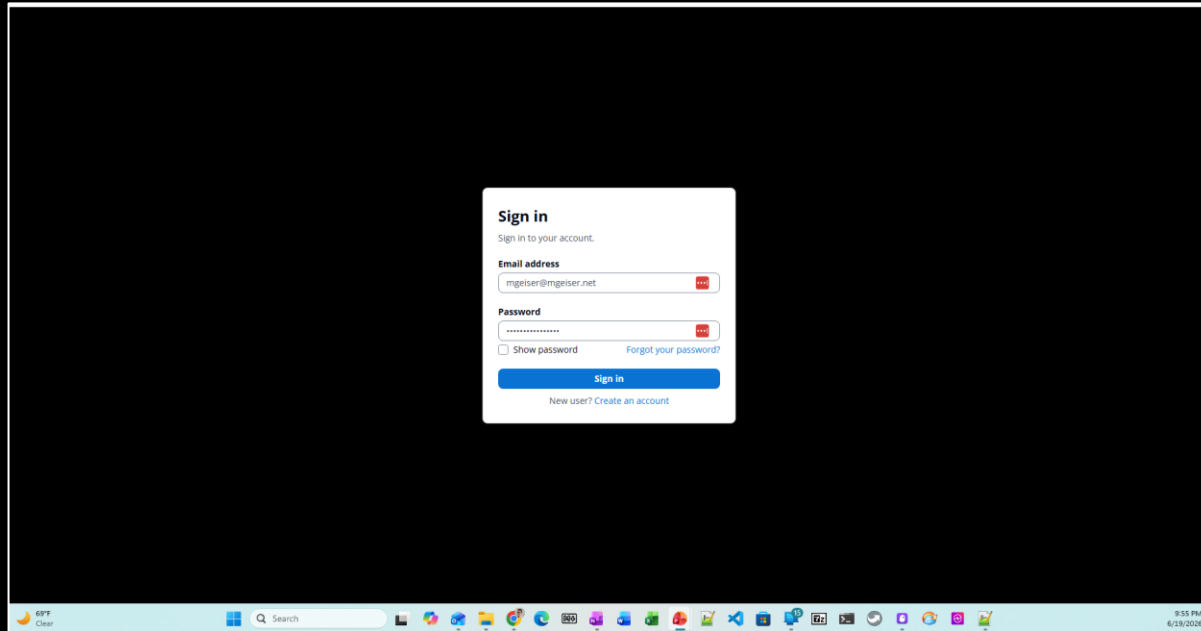


Web



Mobile

Cognito Based User IAM



Groups:

- Users
- Administrators

Other Cognito Functions

- User Signup
- Password reset
- Password recovery
- MFA not implemented in MVP

Audio File Upload

 **Presentation Coach**

[Administrator](#) [Upload](#) [Submissions](#) [Logout](#)

Upload Presentation

Presentation File * Required

Choose File

1_Introducing AgentCore.mp3

1_Introducing AgentCore.mp3 (2.45 MB)

Presentation Title * Required

test run

9 / 200

Description (optional)

Describe your presentation

0 / 2000

Upload

 © 2026 Geislersoft Presentation Coaching Platform

Features

- File picker
- Title
- Description
- Video File upload on roadmap

Processing

Features

- Multiple steps
- User updates

 **Presentation Coach**

Administrator Upload Submissions Logout

My Submissions

test 3

File: 3_Agentcore-Introduction.mp3
AgentCore Introdut from AWS
Uploaded: Jun 20, 2026, 02:37 PM

Pending

Delete

 **Presentation Coach**

Administrator Upload **Submissions** Logout

My Submissions

test 3

File: 3_Agentcore-Introduction.mp3
AgentCore Introdut from AWS
Uploaded: Jun 20, 2026, 02:37 PM

Processing...

Processing

Delete

test of another presentation

File: 2_Andy Jassy AgentCore.mp3
Uploaded: Jun 20, 2026, 11:12 AM

Completed

Completed

Delete

Download Report

test repeat

File: 1_Introducing AgentCore.mp3
Uploaded: Jun 20, 2026, 09:07 AM

Completed

Completed

Delete

Download Report

test

File: 1_Introducing AgentCore.mp3
Uploaded: Jun 19, 2026, 02:52 PM

Failed

Failed

Delete

Ready to Read

 Presentation Coach		Administrator	Upload	Submissions	Logout
My Submissions					
test 3 File: 3_AgentCore-Introduction.mp3 AgentCore Introdut from AWS Uploaded: Jun 20, 2026, 02:37 PM				Completed	
Download Report				Delete	
test of another presentation File: 2_Andy.Jassy.AgentCore.mp3 Uploaded: Jun 20, 2026, 11:12 AM				Completed	
Download Report				Delete	
test repeat File: 1_Introducing AgentCore.mp3 Uploaded: Jun 20, 2026, 09:07 AM				Completed	
Download Report				Delete	
test File: 1_Introducing AgentCore.mp3 Uploaded: Jun 19, 2026, 02:52 PM				Failed	
				Delete	

Features

- Current status of analysis
- Data entered by User displayed
- Download report link
- Delete link

The Report

Presentation Coaching Report

Submission: 3dced911-8c1c-48ff-8374-708a05484d31

Executive Summary

This report covers 7 evaluation dimensions with an overall average score of **6.0/10.0**.

Dimensions evaluated: delivery (6.8), structure (6.2), executive_presence (5.2), technical_communication (7.2), audience_engagement (4.5), pacing (6.2), persuasion (5.8)

Per-Dimension Detailed Feedback

delivery (Score: 6.8/10.0)

Findings:

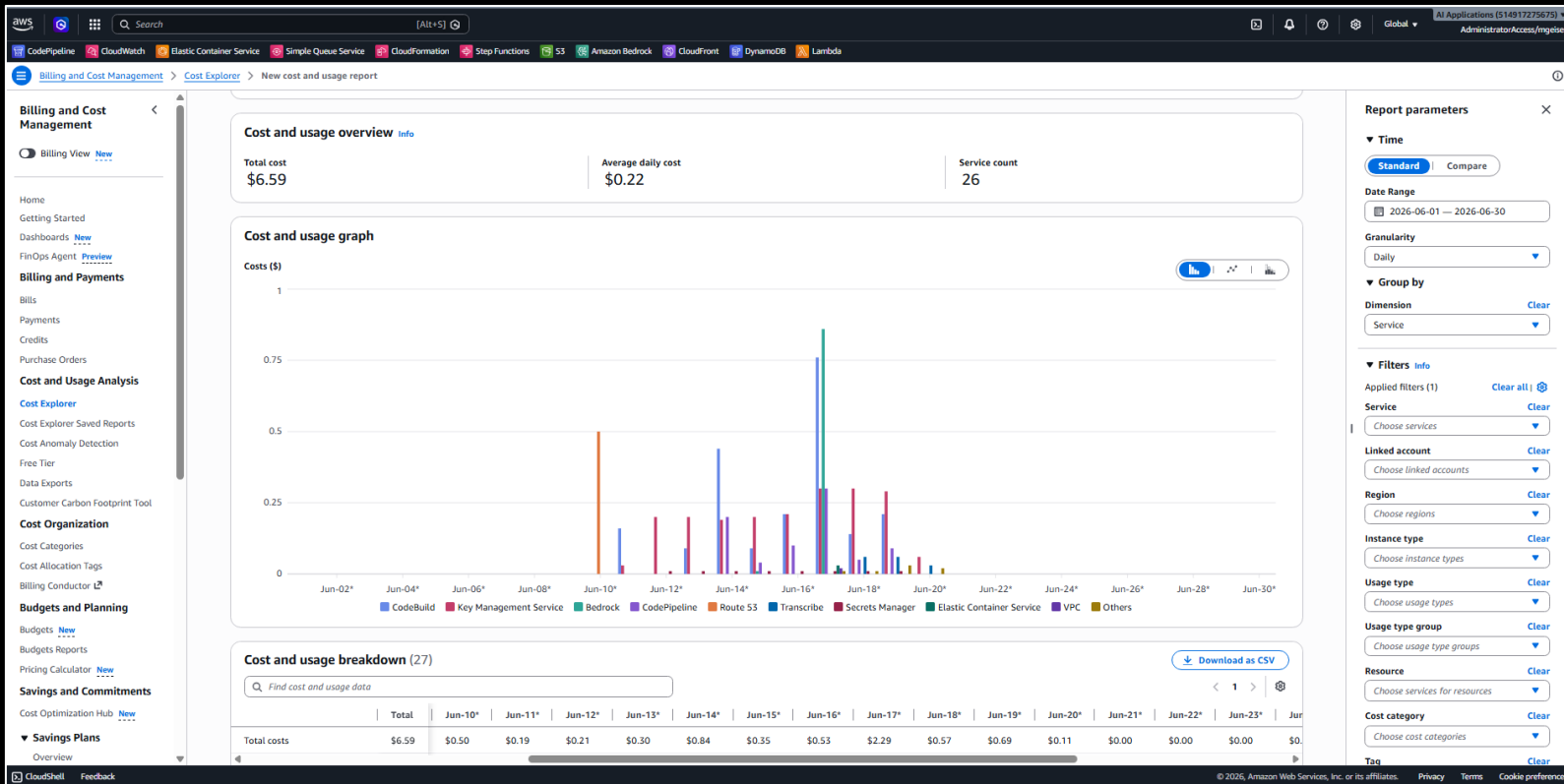
- [LOW] Filler Words: Two distinct 'uh' filler words detected at timestamps 159.455s and 240.789s. While the count is relatively low for a ~5-minute presentation, they occur at moments that slightly interrupt the flow — notably mid-sentence ('It also has built-in, uh, tools for handling memory' and 'The, uh, agent SDKs').
→ Suggestion: Replace 'uh' hesitations with a deliberate silent pause. Practice the transition sentences around memory and agent SDKs specifically, as these appear to be points where recall or phrasing causes momentary uncertainty.
- [MEDIUM] Pace: The overall speaking rate averages approximately 155-165 words per minute across the ~307-second presentation, which is within the acceptable range. However, several dense technical passages — such as the enumeration of frameworks ('OpenAI's agents SDK, LangGraph, LangChain, Crew AI') and the capabilities overview — are delivered at a compressed pace with minimal pausing between items, making them harder to absorb.
→ Suggestion: Slow down deliberately when listing multiple technical items or product names. Insert 1-2 second pauses between listed frameworks and capabilities to allow the audience time to process each item. Consider chunking the list with a slight rising intonation on each item and a clear drop on the final one.
- [MEDIUM] Pauses: The most notable strategic pause occurs between the 'Amazon Technologies' sentence end (~178.7s) and the 'So, at a high level' transition (~180.0s) — approximately a 1.3-second gap, which is effective. Similarly, there's a clean pause after 'Big one is memory' (~221s). However, many sentence boundaries show only 0.1-0.3 second gaps, and transitions between major topic areas (e.g., moving from deployment to the toolkit, from toolkit to capabilities) lack sufficiently strong pauses to signal topic shifts.
→ Suggestion: Add deliberate 2-3 second pauses at major section transitions, especially before 'What is Agent Core?', 'So at a high level, let's look at the capabilities', and 'So let's get started.' These act as

Features

Detailed feedback (10 – 13 Pages) across 7 dimensions:

- Delivery
- Structure
- Executive presence
- Technical communication
- Audience engagement
- Pacing
- Persuasion

Development Costs



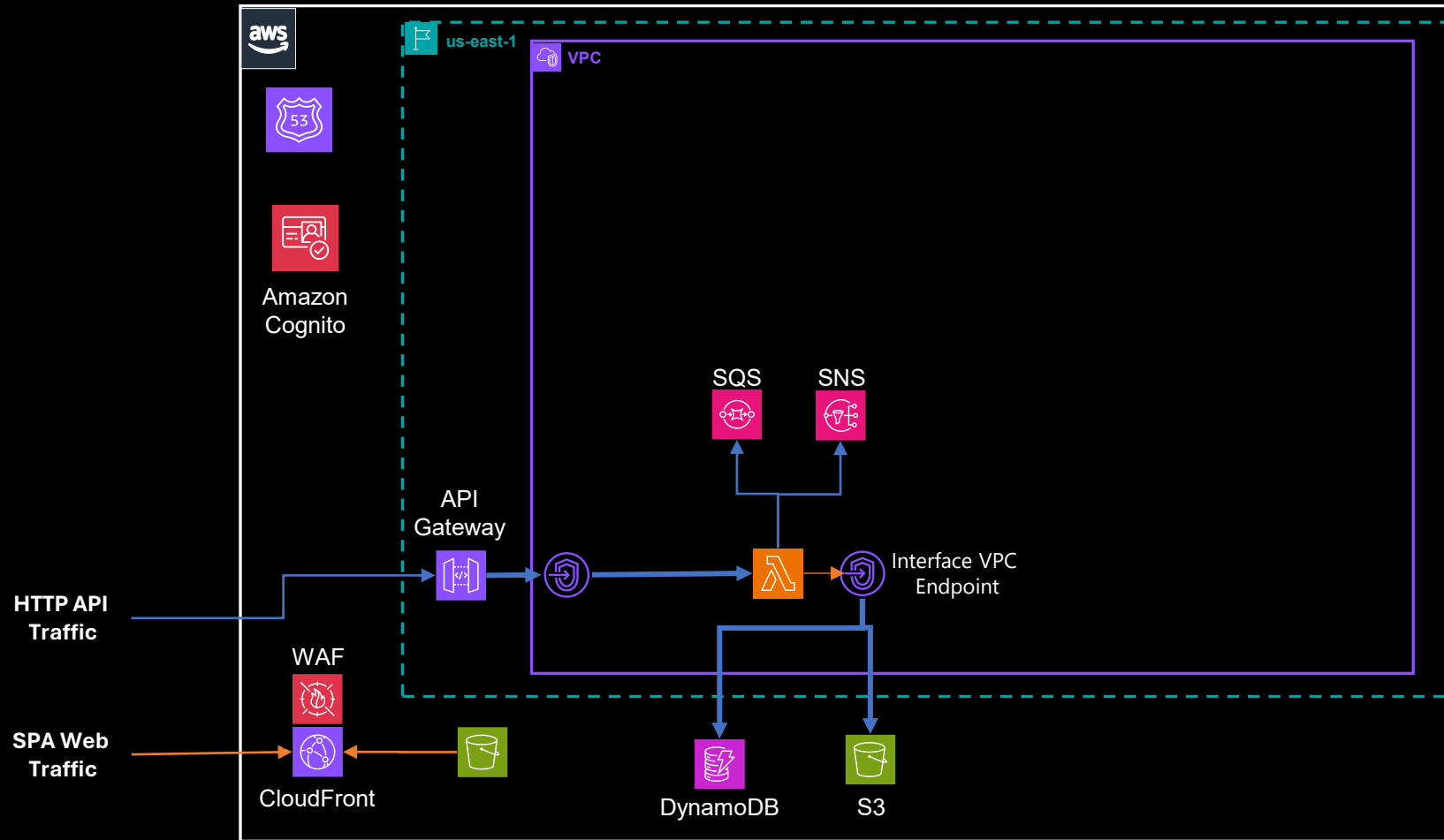
Features

Total costs of the development

- AWS Account: \$6.59
- Kiro AI-DLC: 1355 Credits (\$27.10)

The Architecture

Web App SPA and Backend

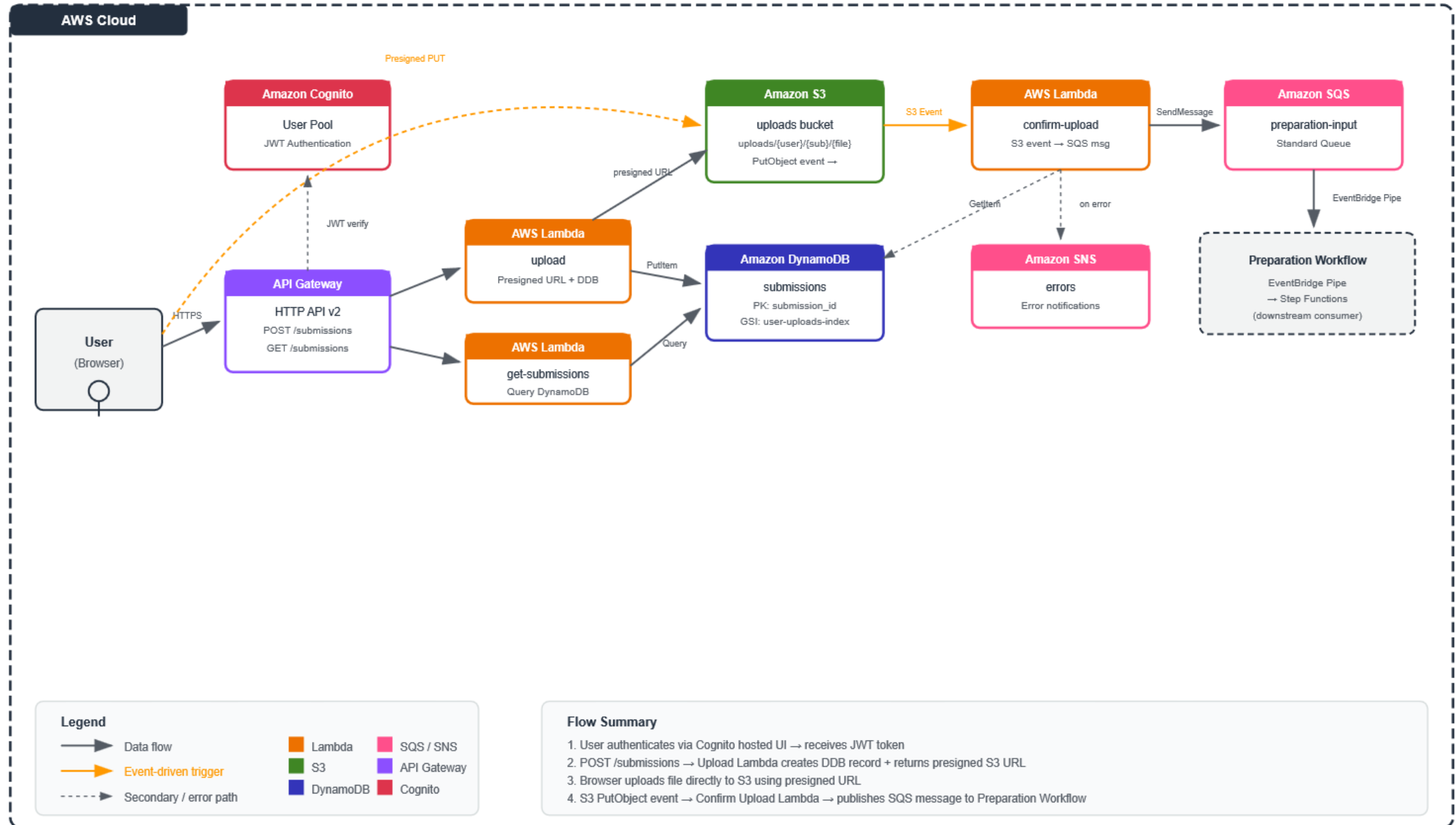


Ask

- Are VPC Endpoints used between API Gateway and Lambdas?
- Are Interface VPC Endpoints used between Lambdas and DynamoDB and S3?
- Are Lambdas fully isolated from AZ failures?
- Where is SQS?
- Where is SNS?

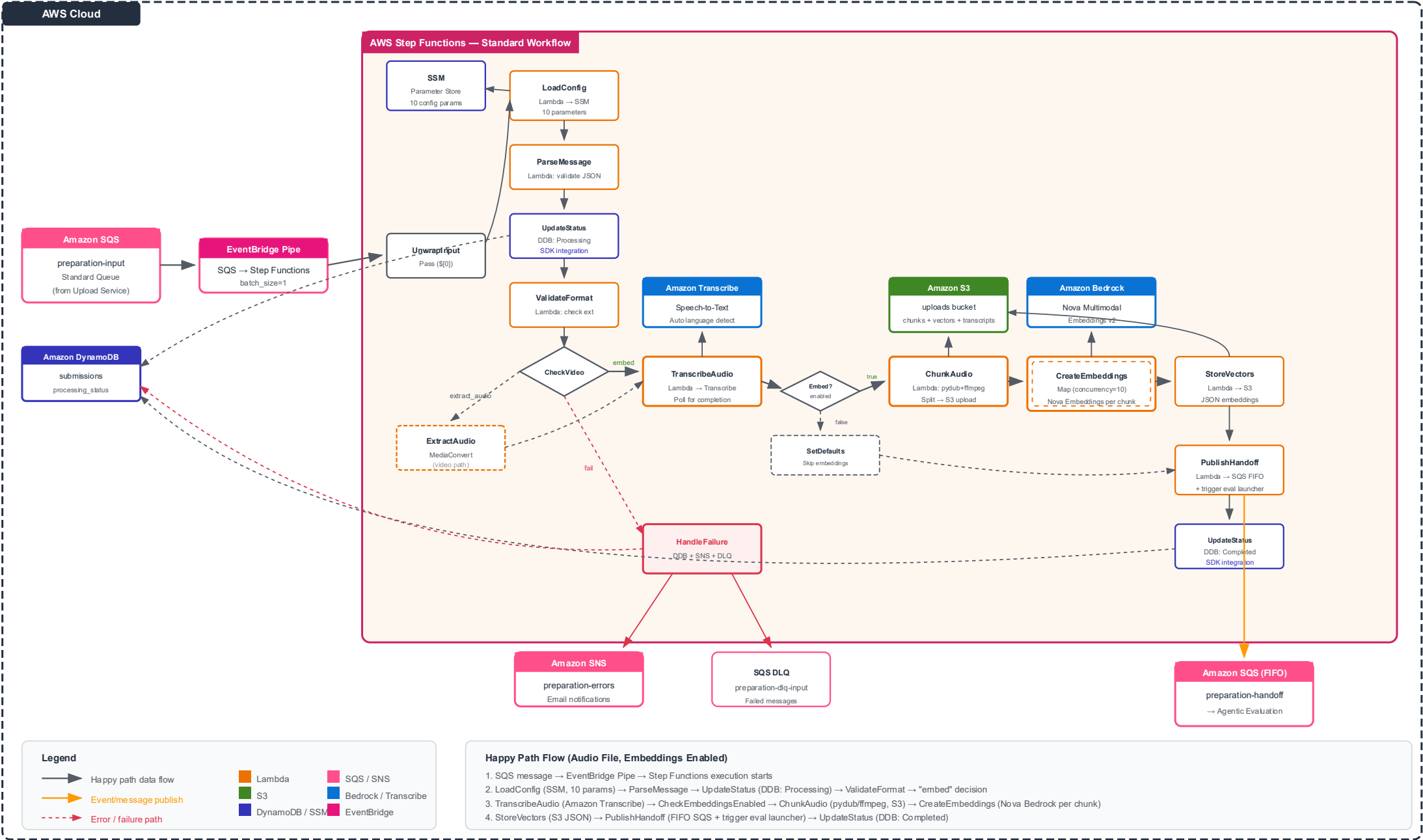
Upload and Storage Service — Architecture Diagram

prescoach-dev-kiro



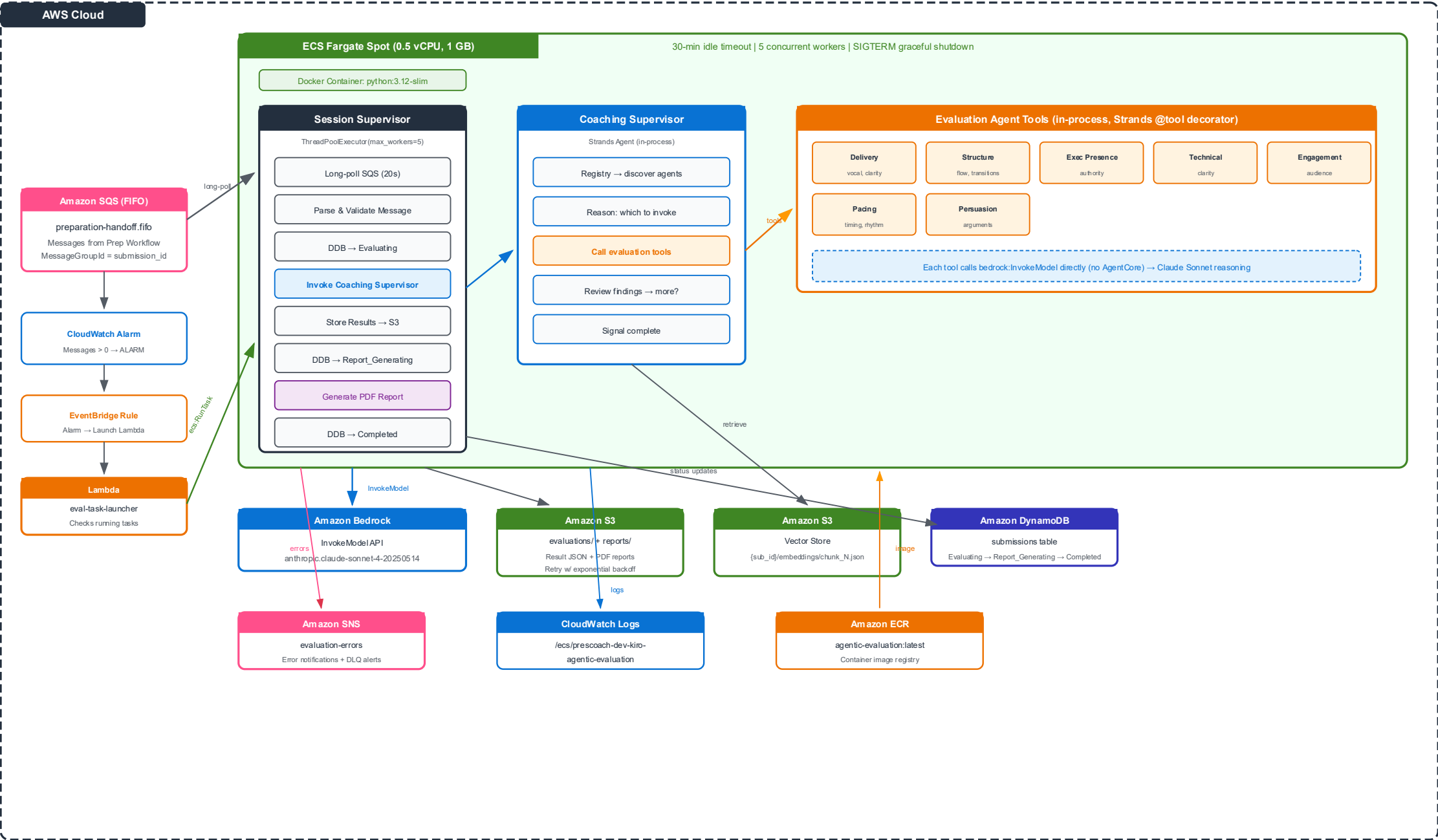
Preparation Workflow — Architecture Diagram

prescoach-dev-kiro | Step Functions Standard Workflow | 10 Lambda Functions



Agentic Evaluation — LOCAL MODE Architecture

ECS Fargate Spot | Strands SDK Local Execution | Direct Bedrock InvokeModel



LOCAL MODE — Key Characteristics

- All agents run in-process inside a single ECS Fargate Spot container (no AgentCore deployment required)
- Strands SDK creates Agent objects locally — each evaluation tool calls bedrock:InvokeModel directly
- Session state is in-memory only (lost when task exits) — no persistent memory across task launches
- Cost-optimized: Fargate Spot (70% discount), 30-min idle timeout, concurrent processing of 5 submissions
- Cold start: ~45-90s (container pull + startup). Kept warm for 30 min after last message.

Cost Estimate

Per evaluation task (5 min): ~\$0.0007

Monthly @ 5/day: ~\$0.10

Monthly @ 100/day: ~\$2.10

(Bedrock model costs are additional)

Agentic Evaluation — AGENTCORE MODE Architecture

Amazon Bedrock AgentCore Runtime | Managed Agents | Session Memory | Auto-Scaling

